

# Liver Metastases: Correlation between Imaging Features and Pathomolecular Environments

Kumi Ozaki, MD, PhD

Shohei Higuchi, MD

Hirohiko Kimura, MD, PhD

Toshifumi Gabata, MD, PhD

**Abbreviations:** ADC = apparent diffusion coefficient, GIST = gastrointestinal stromal tumor, NET = neuroendocrine tumor

**RadioGraphics** 2022; 42:1994–2013

<https://doi.org/10.1148/rg.220056>

**Content Codes:**  

From the Departments of Radiology (K.O., H.K.) and Pathology (S.H.), Faculty of Medical Sciences, University of Fukui, 23-3 Matsuoka-Shimoaizuki, Eiheiji, Fukui 910-1193, Japan; and Department of Radiology, Graduate School of Medical Sciences, Kanazawa University, Kanazawa, Japan (T.G.). Presented as an education exhibit at the 2021 RSNA Annual Meeting. Received March 23, 2022; revision requested June 13; revision received June 18; accepted June 22. For this journal-based SA-CME activity, the authors, editor, and reviewers have disclosed no relevant relationships. **Address correspondence to** K.O. (email: [ozakik-rad@umin.org](mailto:ozakik-rad@umin.org)).

©RSNA, 2022

## SA-CME LEARNING OBJECTIVES

*After completing this journal-based SA-CME activity, participants will be able to:*

- Describe the mechanisms underlying organ-specific liver metastases and the role of tumor exosomes in formation of premetastatic niches.
- Discuss how the desmoplastic reactions produced by cancer-associated fibroblasts explain peripheral and peritumor enhancement of colorectal liver metastases, arteriportal shunts accompanying pancreatic ductal adenocarcinoma metastases, and pseudocirrhosis.
- List the characteristic imaging findings of liver metastases from specific primary cancers.

*See [rsna.org/learning-center-rg](https://rsna.org/learning-center-rg).*

A wide range of imaging manifestations of liver metastases can be encountered, as various primary cancers preferably metastasize to the liver (organ-specific metastases), with the imaging characteristics largely depending on various primary tumor-specific factors such as histopathologic category, degree of tumor differentiation, histologic behavior, and intratumor alterations. Characteristic imaging features potentially can help provide a more precise diagnosis in some clinical settings. These settings include those of (a) primary cancers of hollow organs such as gastrointestinal organs, the lungs, and the bladder, owing to the appearance of metastases that cannot be applied to the liver, which is a parenchymal organ; (b) unknown primary tumors; (c) more than one primary tumor; (d) another emergent malignancy; and (e) transformation to a different histopathologic tumor subtype. The characteristic features include the target sign on T2-weighted MR images or during the hepatobiliary phase of hypovascular metastasis, the peripheral rim washout sign on delayed phase images, peritumor hyperintensity during the hepatobiliary phase, hypervascular metastasis, a cystic appearance with marked hyperintensity on T2-weighted images, marked hyperintensity on T1-weighted images, calcification, capsular retraction, absence of the vessel-penetrating sign, distribution of liver metastases, and rare intraductal forms of metastases. In addition to various factors associated with the primary cancer, desmoplastic reactions around the tumor—which can be observed in adenocarcinomas with peripheral and peritumor enhancement, distinct arteriportal shunts with metastases from pancreatic ductal carcinoma, and pseudocirrhosis—also can affect these findings. The authors review the characteristic imaging findings of liver metastases from various primary cancers, with a focus on the mechanisms that underlie organ-specific liver metastases.

*Online supplemental material is available for this article.*

©RSNA, 2022 • [radiographics.rsna.org](https://radiographics.rsna.org)

## Introduction

The liver is one of the most common sites of metastases, and liver metastases are more common than primary liver cancers (1–3). A large number of malignancies—including cancers of the gastrointestinal system (eg, colorectum, stomach, pancreas, biliary), genitourinary system (eg, kidney, prostate), gynecologic system (eg, ovary, endometrium, cervix), neuroendocrine system, and breast; soft-tissue sarcomas; melanomas; and lymphoma (4)—tend to metastasize to the liver.

As patients with liver metastases have a worse prognosis than patients who do not have them (1), accurate diagnosis is crucial for appropriate management of these secondary cancers. Imaging findings have an important role in facilitating accurate evaluations to detect and characterize metastases. Thus, the number, location,

## TEACHING POINTS

- Studies have shown that the primary tumor antecedently consolidates the microenvironment of the metastatic site (soil), for ease in colonization of tumor cells (seeds), by effusing organ-specific exosomes; this process constitutes the formation of a premetastatic niche.
- Desmoplastic reactions around colorectal liver metastases are closely related to the peripheral and peritumor enhancement seen on arterial phase and portal venous phase images.
- Extremely dense desmoplastic reactions tend to compress the peripheral vessels and reduce blood flow, and more hypotensive portal or hepatic veins are more easily affected. This phenomenon can be depicted as arteriportal shunts accompanied by metastatic nodules.
- Metastatic adenocarcinoma with fibrous stroma occasionally may demonstrate central areas of relative hyperintensity but typically has lower signal intensity compared with the background parenchyma; this is described as the target sign, which is observed in 47%–70% of metastatic adenocarcinomas.
- Transformation of a primary cancer to a different histopathologic type can also affect the imaging findings, reflecting histologic features, with the liver metastasis also exhibiting an appearance similar to that of the primary change.

and size of liver metastases need to be accurately determined. In addition, assuming that there may be cases of more than one primary cancer, new primary cancers that appear unexpectedly, or primary cancers that are present but not detected, it is important to understand the characteristic imaging features of liver metastases from various primary malignancies. This is because treatment strategies can differ according to the given primary cancer. For example, a solitary metastasis from colorectal adenocarcinoma can be resected, while one from pancreatic adenocarcinoma needs to be treated with chemotherapy.

## Frequency of Liver Metastases

Liver metastases have been frequently identified in patients who have secondary tumors from various primary cancers, mainly those from gastrointestinal organs, according to data from the SEER database (1) and knowledge of different histologic subtypes (Table 1). Data from another database limited to adenocarcinoma cases (5) have shown the liver to be one of the organs most affected by metastatic disease, which arises most frequently from primary sites in the colorectum, pancreas, stomach, breast, and lung (Table 2). More than 50%–90% of colorectal, breast, gastric, lung, and prostate cancers are adenocarcinomas (6). The second most frequently observed subtype is squamous cell carcinoma, which is found with several cancers (6). A large number of other malignant tumors

from almost any site—including neuroendocrine tumor (NET), gastrointestinal stromal tumor (GIST), malignant melanoma, soft-tissue sarcoma, and lymphoma—can metastasize to the liver (4).

Thus, a better understanding of the frequent sites of primary cancers could be useful in clinical cases involving more than one primary tumor. For example, in a patient with simultaneous colorectal and prostate carcinomas, multiple liver masses may likely be due to colorectal liver metastases, considering the frequency of these cancers (Table 1).

## Why the Liver Is a Dominant Site of Metastases

Tumor cells are known to colonize in specific organs (organ-specific metastases) other than the liver. For example, 91% of ovarian cancers metastasize to the peritoneum (Table 2) (5), 90% of prostate cancers metastasize to bone, and 85% of renal cell carcinomas metastasize to the lung. Even so, various primary cancers preferably metastasize to the liver, and several factors are speculated to be related to organ-specific metastases of the liver (7).

## Blood Flow Factors (Portal Circulation)

The liver's vast blood supply, which originates from the portal and systemic systems, may provide a fertile environment for metastases. On the basis of pathologic examination of the anatomic structure of the vascular system and the accompanying hemodynamic flow, Ewing (8) pointed out the importance of the associated metastatic dissemination and geometric factors. The high frequency of liver metastases that results from pancreatic adenocarcinoma can be explained by Ewing's mechanical theory.

## Factors That Create a Fertile Environment in the Liver Parenchyma

The fenestra of the hepatic sinusoidal endothelium may facilitate formation of a preferable environment for metastases and penetration of malignant cells into the hepatic parenchyma (microenvironment) (9). From pathologic and clinical perspectives, Paget (10) postulated that metastases are not events due to chance; rather, they occur because some tumor cells ("seeds") grow preferentially in the microenvironment of select organs ("soil"), and metastases occur only when the appropriate seed is implanted in suitable soil. The fact that uveal melanoma specifically has liver metastases, even when the primary site is far from the liver (11), may support the potential role of Paget's seed-soil theory, but it cannot be explained by Ewing's theory.

**Table 1: Prevalence of Synchronous Liver Metastases of Primary Cancers in Various Organs**

| Organ of Primary Cancer | Prevalence of Liver Metastases (%) | No. of Patients with Liver Metastases | Total No. of Patients |
|-------------------------|------------------------------------|---------------------------------------|-----------------------|
| Pancreas                | 77.6                               | 25 225                                | 32 519                |
| Colorectum              | 72.0                               | 30 943                                | 42 934                |
| Biliary tract           | 62.8                               | 3740                                  | 5958                  |
| Small intestine         | 62.2                               | 1894                                  | 5958                  |
| Esophagus               | 48.9                               | 3412                                  | 6972                  |
| Stomach                 | 46.2                               | 5757                                  | 6972                  |
| Ovary                   | 26.3                               | 2285                                  | 8690                  |
| Breast                  | 25.7                               | 5286                                  | 20 557                |
| Skin (melanoma)         | 25.6                               | 1146                                  | 4475                  |
| Urinary bladder         | 24.9                               | 1006                                  | 4033                  |
| Lung                    | 24.0                               | 33 052                                | 137 920               |
| Kidney                  | 22.9                               | 2695                                  | 11 761                |
| Bone and soft tissue    | 18.2                               | 595                                   | 3262                  |
| Uterus                  | 16.6                               | 1000                                  | 1011                  |
| Testis                  | 16.2                               | 275                                   | 6022                  |
| Cervix uteri            | 16.0                               | 423                                   | 2637                  |
| Prostate                | 4.8                                | 836                                   | 17 395                |

Source.—Reference 1.

Note.—Data are from the Surveillance, Epidemiology, and End Results (SEER) database of the National Cancer Institute for the years 2010–2015. Among 2.4 million patients with cancer reported in the SEER database, 5.14% of patients were diagnosed with liver metastasis at the time of diagnosis of the primary cancer. The prevalence of liver metastases was variable, depending on the organ of the primary malignancy.

**Table 2: Percentages of Metastatic Sites for Each Primary Tumor Site (Adenocarcinoma)**

| Organ of Primary Cancer | Organ with Metastasis*  |                         |                         |                    |                   |
|-------------------------|-------------------------|-------------------------|-------------------------|--------------------|-------------------|
| Esophagus               | Liver (52) <sup>†</sup> | Lung (20)               | Peritoneum (11)         | Bone (9)           | Brain (6)         |
| Breast                  | Bone (48)               | Liver (30) <sup>†</sup> | Lung (26)               | Brain (7)          | Pleura (7)        |
| Lung                    | Bone (39)               | Brain (25)              | Liver (16) <sup>†</sup> | Adrenal gland (13) | Pleura (10)       |
| Stomach                 | Peritoneum (52)         | Liver (39) <sup>†</sup> | Lung (5)                | Bone (4)           | Brain (2)         |
| Pancreas                | Liver (85) <sup>†</sup> | Peritoneum (14)         | Lung (12)               | Bone (3)           | Pleura (1)        |
| Kidney                  | Lung (67)               | Bone (35)               | Liver (19) <sup>†</sup> | Adrenal gland (13) | Brain (9)         |
| Ovary                   | Peritoneum (91)         | Pleura (15)             | Liver (13) <sup>†</sup> | Lung (3)           | Brain (1)         |
| Colon                   | Liver (78) <sup>†</sup> | Peritoneum (29)         | Lung (20)               | Bone (3)           | Brain (1)         |
| Rectum                  | Liver (71) <sup>†</sup> | Lung (33)               | Peritoneum (11)         | Bone (7)           | Adrenal gland (2) |

Source.—Reference 5.

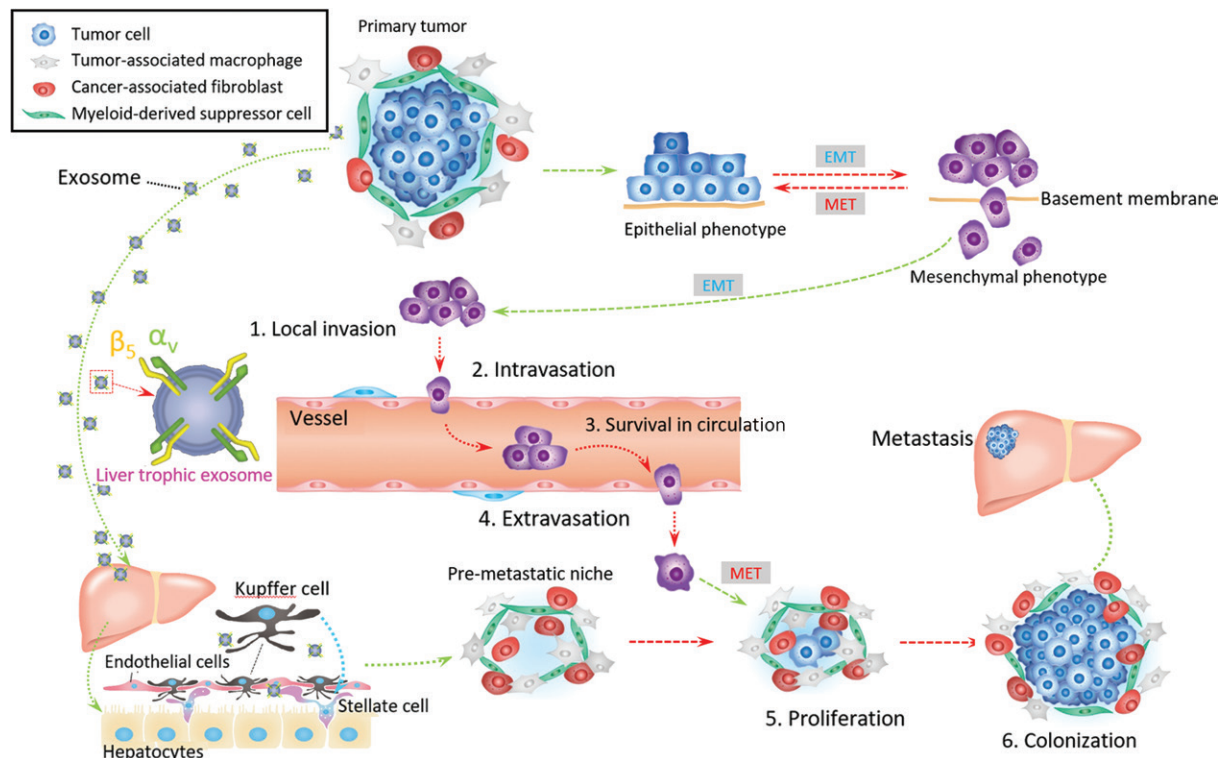
\*The organs harboring metastases from the given primary cancer are shown, with percentage of metastases in parentheses. Data are based on metastases in patients with a confirmed histologic diagnosis of metastatic cancer. The differences in data between Tables 1 and 2 are dependent on the duration and time of year of the data collection, histologic subtypes, and synchronous and/or metachronous liver metastases.

<sup>†</sup>The liver is the organ most frequently affected by metastatic disease.

## Tumor-related Factors

The theories of both Paget (10) and Ewing (8) could be partially correct, but neither theory alone can explain all of the phenomena associated with organ-specific metastases. Studies have shown

that the primary tumor antecedently consolidates the microenvironment of the metastatic site (soil), for ease in colonization of tumor cells (seeds), by effusing organ-specific exosomes; this process constitutes formation of a premetastatic niche (12).



**Figure 1.** Tumor microenvironment and premetastatic niche formed by means of tumor exosome effusion during metastasis progression. The tumor microenvironment is composed of tumor-associated macrophages, cancer-associated fibroblasts, and myeloid-derived suppressor cells in addition to tumor cells. Tumor exosomes derived from the primary cancer are seen as small gray circles. A magnification of one exosome is shown together with integrins  $\alpha_v$  and  $\beta_5$  (left side of figure). The primary tumor antecedently consolidates the microenvironment of the metastatic site by effusing organ-specific exosomes that preferentially fuse with Kupffer cells, which transform hepatic stellate cells into cancer-associated fibroblasts and induce liver fibrosis; this process is formation of a premetastatic niche. Metastatic tumor progression is divided into six major steps: 1. Invasion of the basement membrane and cell migration. 2. Intravasation into the surrounding vasculature or lymphatic system. 3. Survival in the circulation. 4. Extravasation from the vasculature to secondary tissues. 5. Proliferation at the distant location. 6. Colonization at secondary tumor sites. *EMT* = epithelial-to-mesenchymal transition, *MET* = mesenchymal-to-epithelial transition.

Exosomes, which range from 40 to 160 nm in diameter (average, 100 nm), are a subset of extracellular vesicles that are surrounded by a lipid bilayer membrane and secreted by most eukaryotic cells (13). The exosomes play a crucial role in regulating tumor growth, metastasis, and angiogenesis during the process of cancer development (14). It has been shown that different types of proteins (integrins) are found on the surface of exosomes that are derived from different types of malignant cells (15). For example, in the liver, exosomes expressed by the specific integrins  $\alpha_v$  and  $\beta_5$  preferentially fuse with Kupffer cells (15). Thus, tumor exosomes derive organ-specific metastasis, during which there is integrin-mediated fusion with resident cells (16). As cellular changes to the target organ are induced after uptake of the tumor exosomes, the changes can promote proliferation and colonization of cancer cells at the metastatic sites (Fig 1).

### Metastasis Process

Although the mechanisms that drive each step of the metastasis process remain relatively unclear owing to their complexity, recently introduced

engineered models have elucidated key features of the metastatic cascade (17). There are six major steps involved in metastatic progression of tumors: (a) invasion of the basement membrane and cell migration, (b) intravasation within the surrounding vasculature or lymphatic system, (c) survival within the circulation, (d) extravasation from the vasculature to the secondary tissues, (e) subsequent proliferation at distant locations, and finally (f) colonization of secondary tumor sites (Fig 1) (18).

One important cell remodeling process that occurs during the initial local invasion is epithelial-to-mesenchymal transition (19). During this transition, there is fragmentation of the basement membrane along with loss of polarized organization of the epithelial cells, after which the cells become migratory, develop invasive capabilities, and then transform into highly migratory mesenchymal cells. During the reverse process—that is, mesenchymal-to-epithelial transition—this action appears to support a metastatic outgrowth for proliferation at distant sites (Fig 1) (19).



## Imaging Techniques

Accurate detection and characterization of liver metastases depends on optimization of imaging modalities. Multiple imaging techniques including CT, MRI, and fluorine 18–fluorodeoxyglucose PET/CT are widely used. In a meta-analysis (20), MRI, CT, and PET/CT had sensitivity of 93.1% (95% CI: 88.4%, 96.0%), 82.1% (95% CI: 74.0%, 88.1%), and 74.1% (95% CI: 62.1%, 83.3%), respectively. It is important that the majority of the MRI procedures included in the analysis were carried out by using gadoxetic acid (Primovist; Bayer Schering) and included diffusion-weighted imaging.

The hepatobiliary contrast agent gadoxetic acid, which has replaced previously used hepatospecific contrast agents such as mangafodipir trisodium and superparamagnetic iron oxide, has high sensitivity (90.3%) for detecting 1-cm or smaller liver metastases on hepatobiliary phase MR images (21). The diagnostic performance of diffusion-weighted imaging and hepatobiliary phase MRI and the efficacy of combining these sequences for evaluation of liver metastases have been demonstrated in previous studies (22,23).

For example, the per-lesion sensitivity of gadoxetic acid-enhanced MRI combined with diffusion-weighted imaging (90%) was found to exceed that of other imaging modalities, such as CT and PET/CT (sensitivity of 68% and 61%, respectively), especially in detection of lesions smaller than 10 mm (sensitivity of 74% for MRI combined with diffusion-weighted imaging vs 16% and 9% for CT and PET/CT, respectively) (23). In the same report, it was shown that the diagnostic performance of PET/CT was relatively low for detection of small (<10 mm) lesions.

Furthermore, diffusion can be quantified by using the apparent diffusion coefficient (ADC), with low ADC corresponding to restricted diffusion. The mean ADC of malignant lesions has been reported to be  $0.980 \times 10^{-3} \text{ mm}^2/\text{sec}$ , and applying an ADC cutoff of  $1.066 \times 10^{-3} \text{ mm}^2/\text{sec}$  has resulted in specificity of 86.6% and sensitivity of 73.6% for detection of malignancy (24). Although there may be an overlap in the ADC for liver metastases and other benign or malignant focal lesions (24,25), some metastases, such as those of NET and lymphoma, can have characteristically low ADCs (26,27), while others such as mucinous metastases can have higher ADCs (28).

Other MRI sequences, including T1- and T2-weighted imaging with or without fat suppression, as well as CT, also are necessary to characterize focal liver lesions. Thus, CT and MRI findings are reviewed in this article. Gallium 68–tetraazacyclododecane tetraacetic acid-octreotate PET/CT is

increasingly becoming an option for patients with known NET. PET appears to have good sensitivity and specificity (92.9% and 75%, respectively) for detecting liver metastases (29), although further studies are needed to determine whether it should be performed routinely.

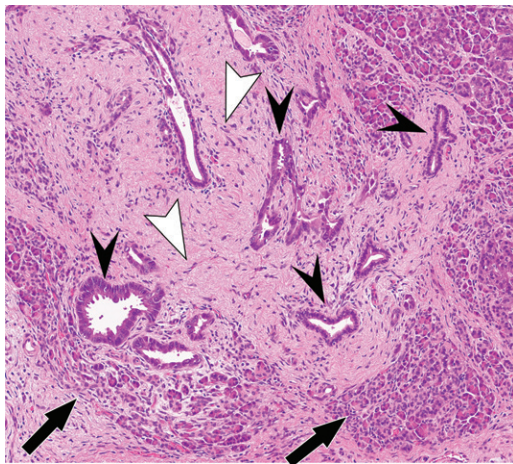
## Vascular Supply of Liver Metastasis and Dynamic Contrast-enhanced Study

For characterization, it is crucial to combine pre- and postcontrast CT and MR images for a dynamic contrast-enhanced imaging study. Solid liver metastases are typically supplied by arterial blood flow (30); hence, they can be classified as hypovascular or hypervascular depending on the degree of the underlying hepatic arterial supply (31). Hypovascular metastases show decreased enhancement relative to the background liver parenchyma and are most conspicuous on portal venous phase images, which are acquired approximately 60–80 seconds after initial injection of contrast agent (32).

Hypervascular metastases enhance earlier than the background liver on arterial phase images, which are acquired approximately 30–40 seconds after the initial injection (32). The lesions show washout on portal phase images and similar or decreased enhancement relative to the background liver parenchyma on delayed phase images, which are acquired approximately 150–180 seconds after the initial injection (32). Thus, dynamic contrast-enhanced images can depict both hypovascular and hypervascular metastases, although the optimal number and choice of acquisition phases for CT remain contentious, given the potential risks associated with higher radiation doses.

## Importance of Noting Specific Imaging Features That Suggest a Specific Primary Cancer

The imaging findings of liver metastases largely depend on various primary tumor-specific factors, such as histopathologic category (eg, adenocarcinoma, squamous cell carcinoma, small cell carcinoma, transitional cell carcinoma, NET, sarcoma, melanoma, lymphoma), tumor differentiation (eg, well, moderate, poor), histologic behavior (eg, solid, cystic, mucinous), and intratumor alterations (eg, necrosis, degeneration, hemorrhage). However, the imaging features of several primary cancers sometimes overlap. For example, liver metastases from colorectal and gastric adenocarcinoma can have similar appearances, manifesting as multifocal hypovascular nodular lesions with hypoattenuation on noncontrast CT images, hypointensity



**Figure 2.** Desmoplastic reaction around the tumor in a 67-year-old man with pancreatic ductal adenocarcinoma. Photomicrograph shows widespread fibrosis and scattered cancer-associated fibroblasts (white arrowheads) around the cancer cells (black arrowheads). Cancer-associated fibroblasts are spindle-shaped cells within the fibrosis. Invasion of the pancreatic parenchyma (arrows) also is observed. (Hematoxylin-eosin stain; original magnification,  $\times 40$ .)

on T1-weighted images, and moderate hyperintensity on T2-weighted images.

Even with overlapping findings, it is not difficult to diagnose liver metastasis in a patient with a known sole primary cancer. However, a better understanding of the typical and characteristic imaging findings could aid in making a more precise diagnosis in the following situations: (a) Primary cancers of hollow organs such as gastrointestinal organs, the lungs, and the bladder, owing to the appearance of metastases that cannot be applied to the liver, which is a parenchymal organ. For such cases, it is important to know the typical findings of each primary cancer. (b) Unknown primary tumors. For example, hypovascular and hypervascular liver masses each would indicate a different primary cancer. (c) More than one primary tumor. For example, a hypovascular mass with calcification would indicate primary colorectal adenocarcinoma rather than primary prostatic adenocarcinoma in a patient with both of these primary cancers. (d) Another emergent malignancy. For example, in a patient with colorectal adenocarcinoma, multiple hypervascular masses might indicate the potential for another primary cancer such as a new NET. (e) Transformation to a different histopathologic subtype. In addition, some characteristic imaging findings represent biomarkers of prognosis or treatment response.

### Desmoplastic Reactions around the Tumor

Desmoplastic reactions associated with malignant neoplasms can evoke a fibrotic response by invading healthy tissue (33). These reactions have been described in association with not only various primary tumors but also metastatic sites (34). Desmoplastic reactions around the tumor, which consist of fibrosis growth caused by cancer-associated fibroblasts around tumor cells (Figs 2, E1), account for part of the formation of the premetastatic niche and can promote local

progression of metastases (Fig 1). These reactions also affect the imaging findings of the metastases and various primary tumor factors. In addition, the desmoplastic reactions produced by cancer-associated fibroblasts, as well as cancer-associated fibroblasts by themselves, might reduce the penetrance of chemotherapy drugs, acting as shields, and contribute to chemoresistance (35).

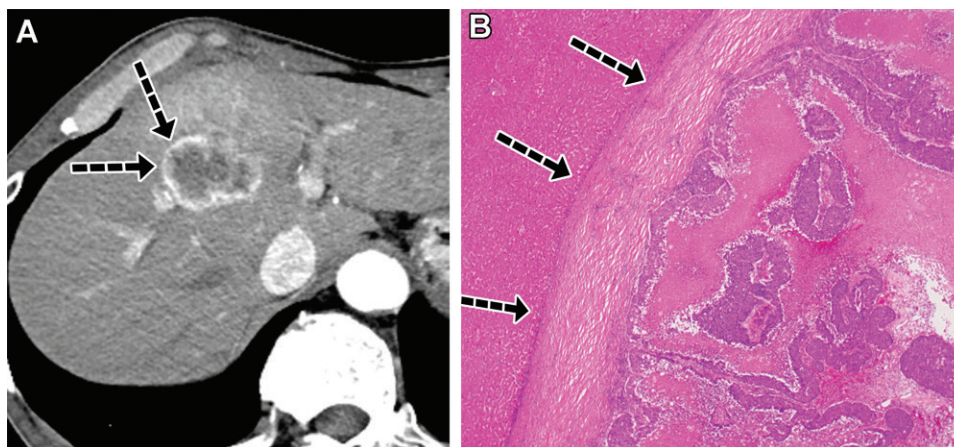
### Peripheral and Peritumor Enhancement

Desmoplastic reactions around colorectal liver metastases are closely related to the peripheral and peritumor enhancement seen on arterial phase and portal venous phase images (Fig 3). Other histopathologic changes in the surrounding hepatic parenchyma, such as lymphocytic infiltration, compressed hepatic parenchyma, and sinusoidal congestion, also can be related to this enhancement (36).

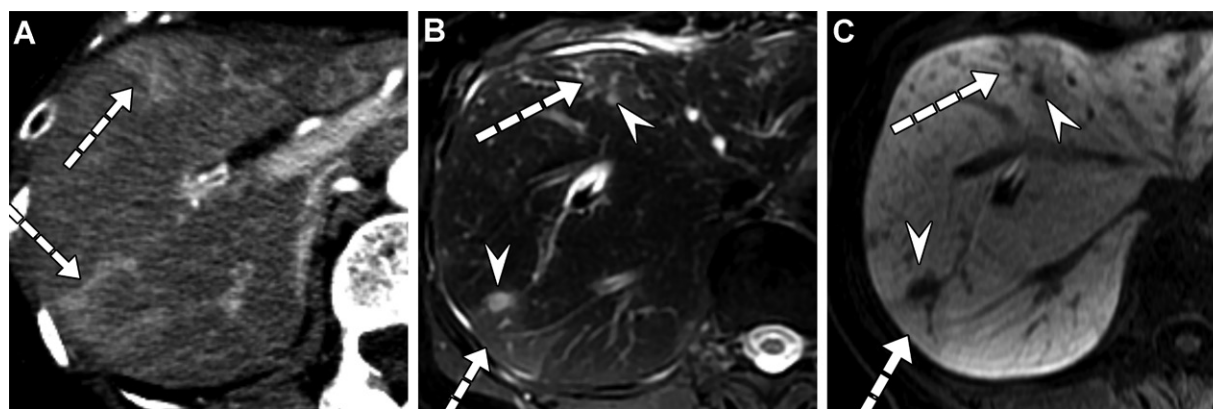
Histopathologically, colorectal liver metastases are classified into three distinct histopathologic growth patterns—desmoplastic, pushing, and replacement—which are based on the interface between the cancer cells of the metastasis and the cancer cells of the surrounding normal liver and different mechanisms to obtain a vascular supply (37). It is important that responses to chemotherapy are different, depending on the histopathologic growth patterns (38). The desmoplastic pattern corresponds to metastases with a prominent desmoplastic reaction, showing peripheral and peritumor enhancement. Thus, the imaging features might enable prediction of the response to chemotherapy, although there have been no studies linking imaging features to histologic subtypes.

### Distinct Arterioportal Shunts with Pancreatic Metastases

Pancreatic ductal adenocarcinoma is the most common form of pancreatic cancer and is characterized by abundant fibrosis, which is caused by desmoplastic reaction around the tumor. Extremely dense desmoplastic reactions are equally observed at the metastatic site, with the fibrous stroma more frequently and excessively broad beyond the cancer cells (34). Extremely dense desmoplastic reactions tend to compress the peripheral vessels and reduce blood flow, and more hypotensive portal or hepatic veins are more easily affected. This phenomenon can be depicted as arterioportal shunts accompanied



**Figure 3.** Peripheral and peritumor enhancement of liver metastasis in a 63-year-old man who underwent resection of sigmoid colon adenocarcinoma 6 years earlier. Axial arterial phase CT image (A) shows a hypovascular metastasis with peripheral and peritumor enhancement (arrows in A), which corresponds to the mainly desmoplastic reaction (arrows in B) in the peritumor area seen on the histologic specimen (B). (Hematoxylin-eosin stain; original magnification,  $\times 10$ .)



**Figure 4.** Arterioportal shunts associated with multiple liver metastases in a 75-year-old man with pancreatic adenocarcinoma. (A) Axial arterial phase CT image shows a triangular region of hyperenhancement (arrows) without a definable lesion. (B) Axial fat-suppressed T2-weighted image shows definable hyperintense masses (arrowheads) at the apex of perilesion hyperintense regions (arrows). (C) Axial hepatobiliary phase MR image shows multiple hypointense metastases (arrowheads), with hypointensity of the associated arterioportal shunts (arrows) due to the desmoplastic reactions that result in reduced portal venous blood flow.

by metastatic nodules (Fig 4). If arterioportal shunts are observed in patients with pancreatic ductal adenocarcinoma, the existence of tiny liver metastases should be assumed.

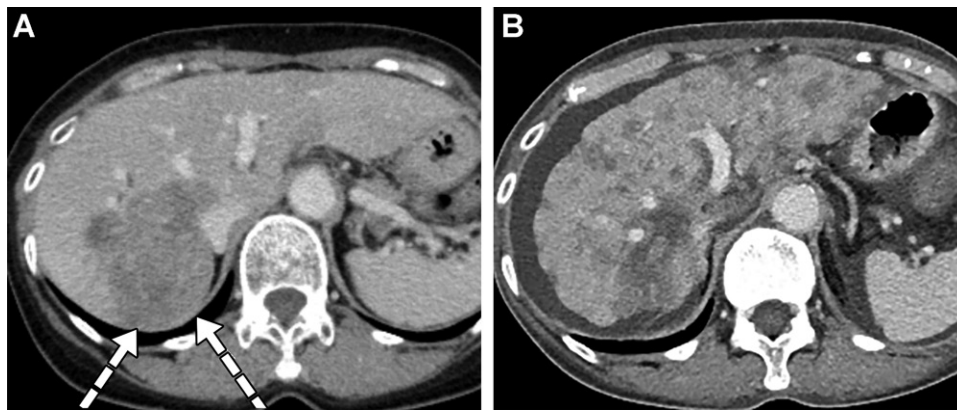
### Pseudocirrhosis

After chemotherapy, pseudocirrhosis, a rare condition, can lead to morphologic changes such as multiple liver metastases. This is seen particularly with liver metastases from breast cancer; however, it is rarely observed with liver metastases from pancreatic, colon, medullary thyroid, or esophageal cancers (39). It is difficult to predict the occurrence of pseudocirrhosis because it is not related to the number of metastases, the histopathologic features, or the chemotherapeutic regimen (40). Although the pathophysiologic mechanism of pseudocirrhosis has yet to be elucidated, it might be deeply involved with desmoplastic

reaction. Therefore, abundant fibrosis caused by desmoplastic reaction around the tumor, fibrotic changes caused by chemotherapy, repeated tumor shrinkage and enlargement, and nodular regenerative hyperplasia in response to chemotherapy might invoke the irregular atrophy-hypertrophy complex of hepatic parenchyma (41).

In addition, a relatively good response to neoadjuvant chemotherapy and a relatively long survival period could lead to morphometric changes in the liver (Fig 5). Although CT images depict hepatic morphometric changes that mimic cirrhosis, residual metastases cannot be separately represented by fibrous changes (41). While ADC-based quantification might facilitate accurate assessment of the hepatic condition by means of discrimination between residual metastases and fibrous change (42), there have been few descriptions of MRI findings.





**Figure 5.** Pseudocirrhosis in a 61-year-old woman with a history of invasive ductal carcinoma of the breast. The patient had undergone repeated chemotherapy with shifting multiagent regimens for liver metastases. (A) Axial CT image 5 years before equilibrium phase CT was performed shows a pretreatment liver metastasis (arrows) in the posterior segment. Subsequently, the posterior metastasis decreased in size, with multiple other emergent metastases repeatedly increasing and decreasing in size for multiple years (not shown). (B) Current axial portal phase CT image shows multiple ill-defined and irregular non-enhancing areas and deformity of the liver, so-called pseudocirrhosis.

**Table 3: Characteristic Imaging Findings of Nodular Lesions**

| Imaging Finding                                                      | Lesions                                                                                                                                       |
|----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Target sign on T2-weighted images of hypovascular metastases         | Colorectal and breast adenocarcinomas                                                                                                         |
| Cystic appearance on T2-weighted images:                             |                                                                                                                                               |
| Severe necrosis or degeneration                                      | NET, GIST, sarcoma, melanoma, angiosarcoma, testicular carcinoma, squamous cell carcinoma                                                     |
| Cystic components of primary tumor                                   | Intraductal papillary mucinous carcinoma of pancreas, adenoid cystic carcinoma, ovarian serous and mucinous carcinomas                        |
| Primary histologic features                                          | Mucinous adenocarcinoma of colorectum and ovary, endometrial carcinoma                                                                        |
| Peripheral rim washout sign                                          | NET; colorectal, breast, and gastric carcinomas                                                                                               |
| Intratumor hemorrhage (fluid-fluid level)                            | NET, GIST, squamous cell carcinoma, melanoma, choriocarcinoma, testicular carcinoma; this finding can be related to necrosis and degeneration |
| Target sign on hepatobiliary phase images of hypovascular metastases | Colorectal and breast adenocarcinomas                                                                                                         |
| Marked hyperintensity on T1-weighted images (melanin)                | Melanoma                                                                                                                                      |
| Peritumor hyperintensity on hepatobiliary phase images               | NET, GIST, colorectal carcinoma                                                                                                               |
| Calcification                                                        | Mucinous adenocarcinoma of colon and ovary, medullary thyroid carcinoma                                                                       |
| Hypervascular metastases                                             | NET, renal cell carcinoma, thyroid carcinoma, choriocarcinoma, pheochromocytoma, malignant melanoma, sarcoma                                  |
| Capsular retraction                                                  | Adenocarcinomas with abundant fibrous stroma, such as lung, breast, and colorectal adenocarcinomas                                            |
| Vessel-penetrating sign                                              | Lymphoma                                                                                                                                      |

### Characteristic Imaging Findings

In diagnosis of liver metastases, several characteristic imaging findings need to be taken into consideration. These findings include tumor vascularity, the characteristic attenuation values and signal intensities of lesions due to specific

primary cancers, specific signs due to characteristic pathologic entities, the pathognomonic distribution, and rare intraductal metastatic forms (Tables 3, 4). Along with accurate detection of the lesion, consideration of the following issues is necessary: whether the imaging findings are



consistent with the known primary cancer, the possibility of another emergent malignancy, and the possibility of transformation into a different histopathologic subtype.

### Target Sign on T2-weighted Images of Hypovascular Metastases

Metastatic adenocarcinomas are the most common type of carcinoma (6) and are often accompanied by various degrees of fibrosis, liquefactive necrosis, and degeneration in predominantly central areas (43). At CT, metastatic adenocarcinomas appear as nonspecific hypovascular lesions with peripheral and peritumor enhancement on arterial or portal venous phase images (Fig 3) (36). In contrast, at MRI, in predominantly central areas, fibrous change or old hemorrhage is depicted as hypointensity on T1- and T2-weighted images, with liquefactive necrosis or degeneration depicted as clear areas of hyperintensity on T2-weighted images and slight hyperintensity on T1-weighted images (44,45).

However, peripherally dominant viable tumor cells are depicted as moderate-hyperintensity lesions on T2-weighted images and hypointense lesions on T1-weighted images (Figs 6, 7, E2A). The histopathologic contrast between central and peripheral areas is seen as the target sign on T2-weighted images (Figs 6B, 7A, E2A) (44). This finding has been observed in 25% of metastases, those from colorectal adenocarcinoma in particular (46).

### Peripheral Rim Washout Sign

The peripheral washout sign is defined as an enhancing liver lesion with a peripheral rim of decreased enhancement relative to its center and the surrounding liver parenchyma on delayed phase MR images obtained by using extracellular contrast agents (Fig E3) (47). The washout sign is observable in 25% of patients with liver metastases from NET or from colorectal, breast, or gastric carcinomas (47). Peripheral washout may reflect a transitory perfusion within the well-vascularized peripheral portion of the lesion, which has both good arterial supply and good venous withdrawal of contrast material (47).

### Target Sign on Hepatobiliary Phase Images of Hypovascular Metastases

Metastases are usually hypointense on hepatobiliary phase images because of their lack of normal hepatocytes. However, metastatic adenocarcinoma with fibrous stroma occasionally may demonstrate central areas of relative hyperintensity but typically has lower signal intensity compared with the background parenchyma; this is described as the target sign, which is observed

**Table 4: Pathognomonic Distribution and Intraductal Metastatic Forms**

#### Pathognomonic distribution

Diffuse small nodular metastases (miliary liver metastases): lung and breast carcinomas, uveal melanoma, NET

Prominently located in gallbladder fossa: malignant gallbladder tumors

#### Intraductal forms

Intrasinusoidal metastases: breast, small cell lung, and gastric carcinomas; melanoma

Portal vein tumor thrombosis: gastric and colorectal adenocarcinoma, gastric NET

Biliary metastases (including extrahepatic bile duct):

Biliary epithelium metastases: colorectal, gastric, breast, uterine, and ovarian carcinomas

Intraluminal metastases: colorectal carcinoma, melanoma, hepatocellular carcinoma

Lymphatic invasion: gastric carcinoma, lymphoma

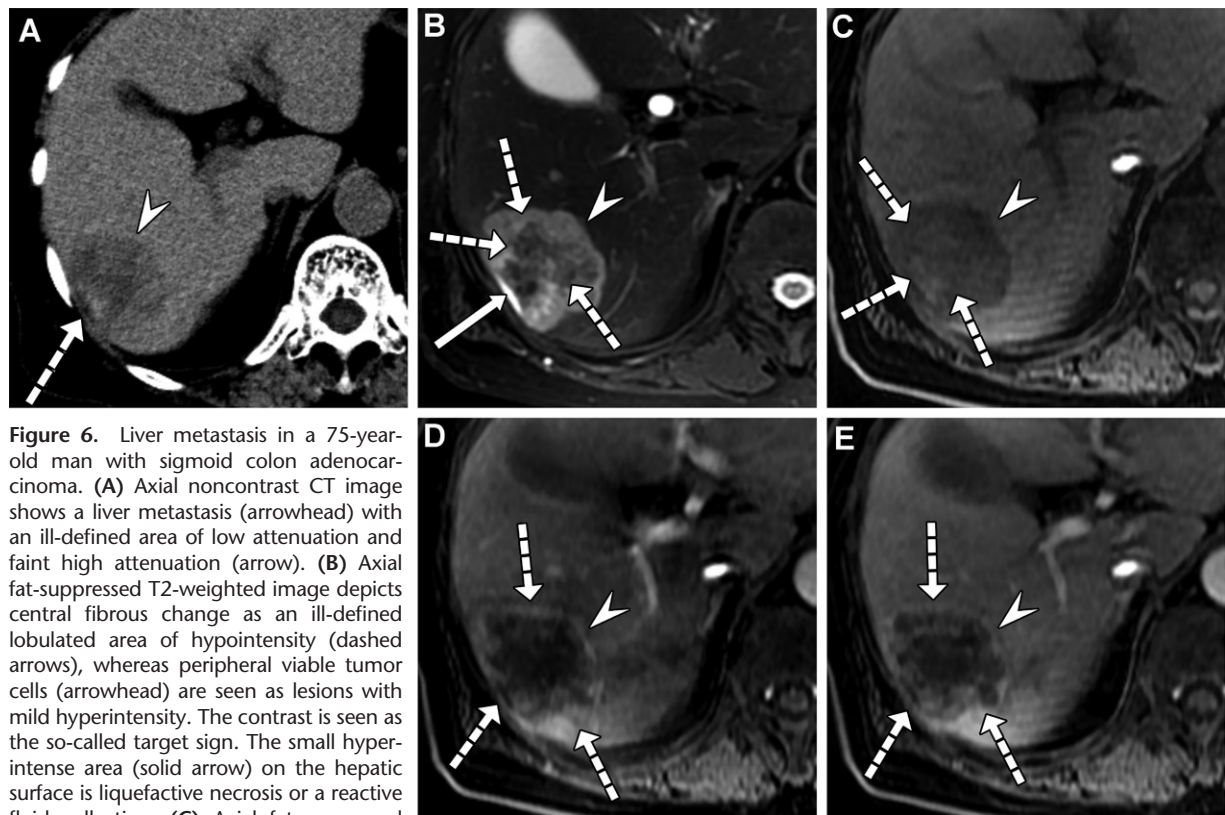
Note.—Multifocal nodular lesions represent the most common distribution of liver metastases, but two characteristic distributions should be considered. Most metastases form nodular lesions in the liver parenchyma and rarely appear as ductal metastases.

in 47%–70% of metastatic adenocarcinomas (Figs 7C, E2B) (48,49). While either increased uptake due to retention in fibrotic stromata or aberrant expression of organic anion-transporting polypeptide 1B3 has been suggested as the mechanism underlying this finding, this remains a topic of debate (50). The target sign of metastases of breast cancer may be associated with a poor response to chemotherapy (51).

### Peritumor Hyperintensity on Hepatobiliary Phase Images

With some liver metastases, such as from NET, GIST, or colorectal adenocarcinomas (Fig E4) (49), peritumor hyperintensity, which is a homogeneously hyperintense rim surrounding the tumor, can be observed on hepatobiliary phase images, although the frequency of this finding has not been established. Histopathologically, peritumor hyperintensity corresponds to peritumor hyperplasia, which is defined as a rim of hyperplastic hepatocytes surrounding the tumor (52).

The pathogenesis of peritumor hyperplasia remains controversial, but two mechanisms should be considered. One possibility is that a perfusion abnormality due to tumor vascular invasion causes increased arterial blood flow in the peritumor liver parenchyma and consequent hepatocyte hyperplasia. Another possibility is



**Figure 6.** Liver metastasis in a 75-year-old man with sigmoid colon adenocarcinoma. (A) Axial noncontrast CT image shows a liver metastasis (arrowhead) with an ill-defined area of low attenuation and faint high attenuation (arrow). (B) Axial fat-suppressed T2-weighted image depicts central fibrous change as an ill-defined lobulated area of hypointensity (dashed arrows), whereas peripheral viable tumor cells (arrowhead) are seen as lesions with mild hyperintensity. The contrast is seen as the so-called target sign. The small hyperintense area (solid arrow) on the hepatic surface is liquefactive necrosis or a reactive fluid collection. (C) Axial fat-suppressed T1-weighted image shows a slightly hyperintense area (arrows), indicating necrosis or a small amount of hemorrhage within the metastasis (arrowhead). (D) Axial arterial phase MR image shows peripherally dominant enhancement (arrows) and that the entire nodule (arrowhead) is hypovascular. (E) Axial portal phase MR image shows that the peripherally dominant enhancement (arrows) is prolonged, and the entire nodule (arrowhead) is still hypovascular.

that regenerative changes occur in hepatocytes that are compressed by tumors with expansive growth (49). A relationship between degree of peritumor hyperintensity and progression-free survival has been observed with liver metastases from NET (53).

### Hypervascular Metastases

Hypervascular metastases typically arise from hypervascular primary tumors, including NET, renal cell carcinoma, thyroid carcinoma, choriocarcinoma, pheochromocytoma, malignant melanoma, and sarcomas (54), which show hypoattenuation on nonenhanced CT images, moderate hypointensity on T1-weighted images, and moderate hyperintensity on T2-weighted images. Dynamic contrast-enhanced CT and MRI depict early enhancement on arterial phase images and washout on portal venous phase images (Fig 8) (32), which are indistinguishable from the enhancement findings of hepatocellular cancer (Fig E5).

However, multifocal hypervascular lesions in patients without cirrhosis might be more likely to be liver metastases (55). In addition, liver metastases from gastrointestinal or breast adenocarcinoma potentially can have an appearance

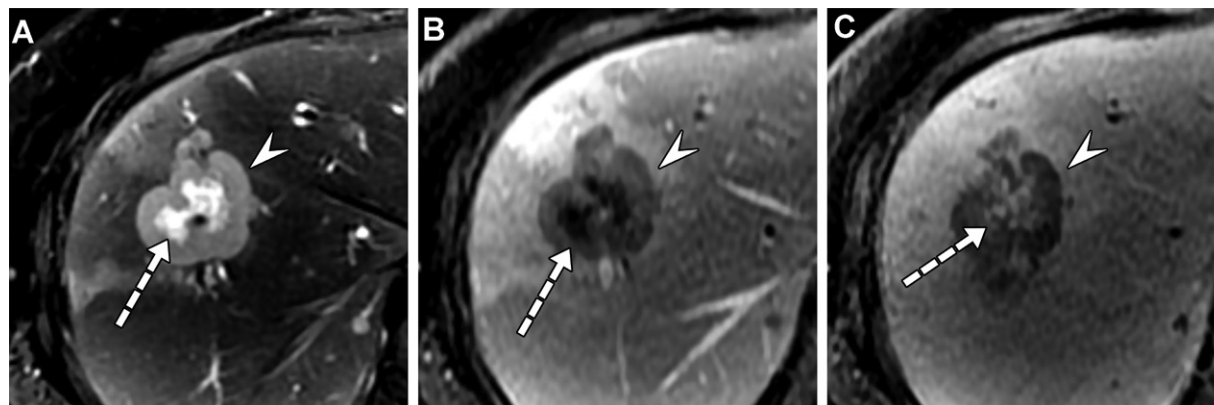
similar to that of a hypervascular lesion owing to marked peripheral and peritumor enhancement (Fig E6) (36,54).

The washout in small ( $\leq 5$  mm) metastases from NET on portal venous phase images is sometimes unclear, although arterial phase hyperenhancement can be observed (56). In addition, hypervascular metastases frequently accompany various degrees of degeneration (Fig 9) (54,57).

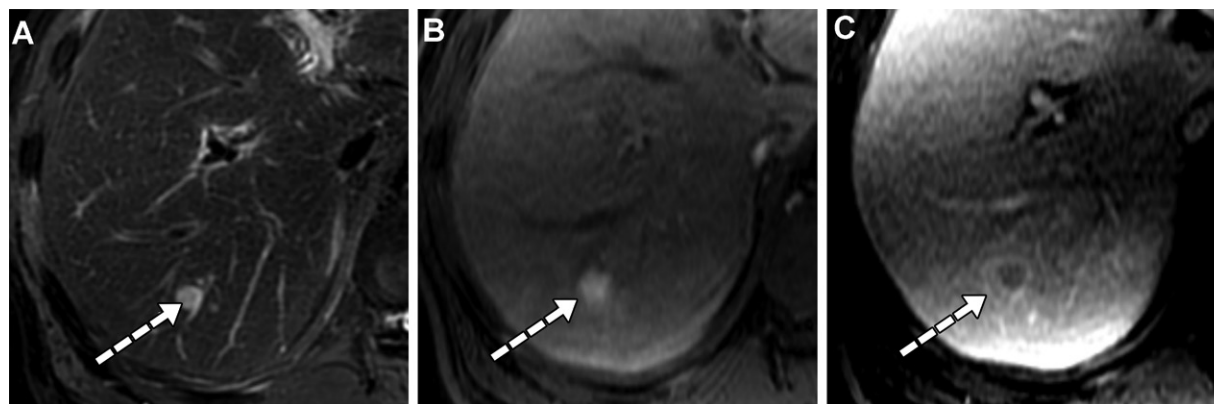
### Cystic Appearance on T2-weighted Images

Cystic appearances include the following three entities: cystic changes due to necrosis or degeneration, cystic metastases due to the cystic components of the primary tumor, and marked hyperintensity on T2-weighted images due to the primary histologic features. These three entities have the potential to appear as the same finding, showing marked hyperintensity at T2-weighted MRI, and can be indistinguishable from each other. Thus, these three conditions have sometimes been comprehensively referred to as cystic liver metastases (57–59).

Severe necrosis or degeneration tends to appear in metastases from NET, GIST, sarcoma, melanoma, angiosarcoma, testicular

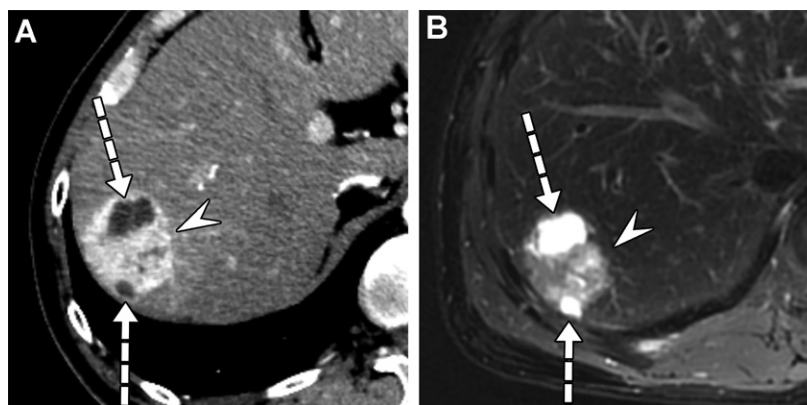


**Figure 7.** Synchronous liver metastasis in a 52-year-old woman with invasive ductal carcinoma of the breast. (A) Axial fat-suppressed T2-weighted image shows central liquefactive necrosis (arrow), depicted as an area of clear hyperintensity, whereas peripheral viable tumor (arrowhead) is depicted as an area of moderate hyperintensity. The contrast between the peripheral and central areas forms the so-called target sign. (B) Axial portal phase MR image shows a moderately enhanced peripheral viable lesion (arrowhead), whereas the central liquefactive necrosis (arrow) shows no enhancement. (C) Axial hepatobiliary phase MR image shows central liquefactive necrosis with slight hyperintensity (arrow) in contrast to a clear hypointense peripheral area (arrowhead); these findings form the target sign.



**Figure 8.** Hypervascular metastasis in a 77-year-old man with a history of grade 2 pancreatic NET 3 years previously. (A) Axial fat-suppressed T2-weighted image shows a hyperintense liver metastasis (arrow). (B) Axial arterial phase MR image shows early enhancement of the metastasis (arrow). (C) Axial portal phase MR image shows washout from the metastasis (arrow). The background liver image findings were not indicative of chronic liver disease.

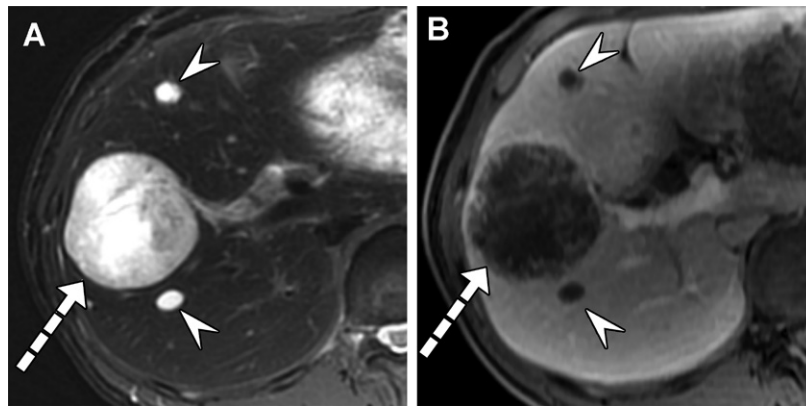
**Figure 9.** Hypervascular metastasis with liquefactive necrosis in a 41-year-old woman with a history of grade 2 pancreatic NET 3 years previously. (A) Axial arterial phase CT image shows a hypervascular metastasis (arrowhead) with nonenhancing areas (arrows), which histologically correspond to liquefactive necrosis. (B) Axial fat-suppressed T2-weighted image shows a viable lesion (arrowhead), which is depicted as a lesion with moderate hyperintensity, whereas the liquefactive necrosis (arrows) is depicted as areas of clear hyperintensity.



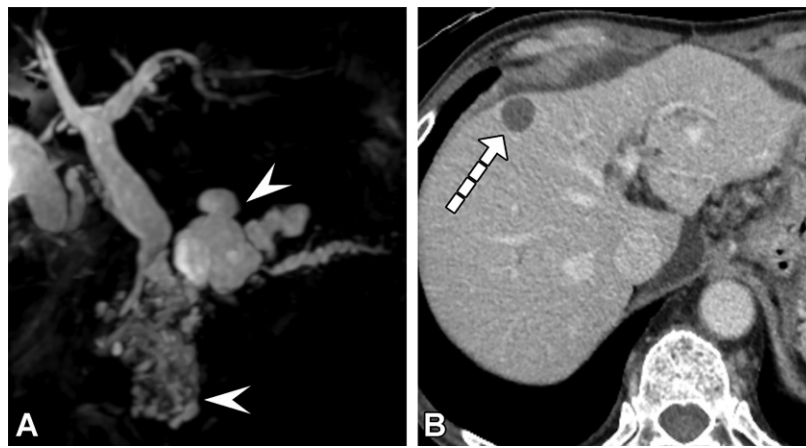
carcinoma, or squamous cell carcinoma from any site (54,57). Rapid growth of hypervascular metastases beyond the blood supply sometimes results in severe degeneration or necrosis and is depicted as characteristic areas of clear

hyperintensity on T2-weighted images and as hypoattenuating areas on CT images (Fig 9). The wall and inner surface of the lesions show ragged, serrated, and ill-defined enhancement due to residual viable tumor cells with fibrotic





**Figure 10.** Liver metastasis with central liquefactive necrosis in a 55-year-old woman who underwent resection of a gastric GIST 1½ years previously. (A) Axial fat-suppressed T2-weighted image shows a liver metastasis (arrow) with inhomogeneous hyperintensity and moderate hypointensity. In contrast, simple hepatic cysts (arrowheads) are depicted as clear homogeneous areas of hyperintensity. (B) Axial portal phase MR image shows peripherally dominant irregular enhancement (arrow), which corresponds to scattered viable cells. In contrast, the simple hepatic cysts (arrowheads) show no enhancement.



**Figure 11.** Cystic metastasis due to cystic components of the primary tumor in a 61-year-old man who underwent resection of pancreatic intraductal papillary mucinous carcinoma 2 years earlier. (A) Pre-operative MR cholangiopancreatography image clearly shows intraductal papillary mucinous carcinoma, including multiple cystic components (arrowheads). (B) Axial equilibrium phase CT image shows a clear hypoattenuating lesion (arrow) with no enhancement, which was histologically proven to be metastasis (not shown).

tissues (Fig 10). This irregular enhancement usually can be distinguished from the appearance of a simple hepatic cyst. However, complete necrosis or degeneration of metastases can sometimes have an appearance similar to that of a simple hepatic cyst (Fig E7).

When the primary tumor—for example, intraductal papillary mucinous carcinoma of the pancreas, adenoid cystic carcinoma, or ovarian serous and mucinous carcinoma—consists of cystic components (58), the liver metastases also can be composed of cystic components. These components appear as a multilocular cystic pattern with enhancing septa and irregular thickened walls (Fig E8). In the case of unilocular cystic metastases with limited enhancing walls, the appearance is similar to that of simple hepatic cysts, which exhibit hypoattenuation on noncontrast CT images, clear hyperintensity on T2-weighted images, and hypointensity on T1-weighted images, accompanied by no enhancement (Fig 11).

Liver metastases of mucinous adenocarcinoma or uterine endometrial carcinoma show marked hyperintensity on T2-weighted images owing to abundant mucin or endometrial components, despite lack of necrosis, lack of degeneration,

and lack of cystic components (59). These markedly hyperintense metastases seen at T2-weighted imaging exhibit thick irregularly enhancing walls accompanying internal septa and mural nodules that result from scattering tumor cells (Fig 12).

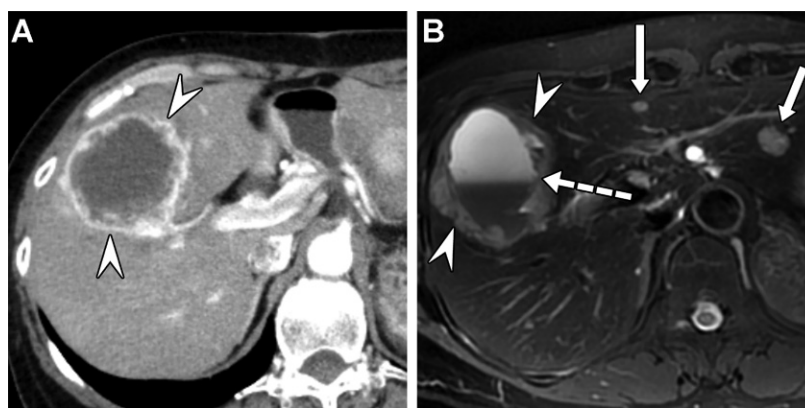
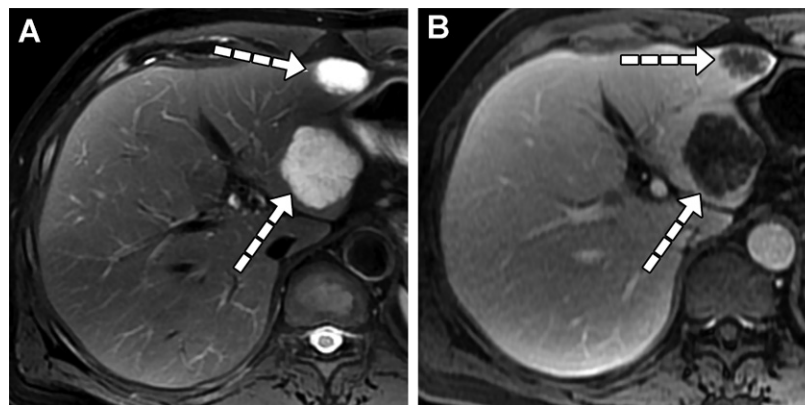
### Intratumor Hemorrhage (Fluid-Fluid Level)

Intratumor hemorrhage (fluid-fluid level) can be caused by the hemorrhagic content, which also is closely related to necrosis or degeneration associated with metastases. This is encountered especially in metastases from NET and is also found in tumors such as melanoma, angiosarcoma, ovarian carcinoma, leiomyosarcoma, and squamous cell carcinoma (60,61). Intratumor hemorrhage can exhibit slight hyperintensity on T1-weighted images (45) and characteristic fluid-fluid levels on T2-weighted images (Fig 13) (60). The attenuation and signal intensity at imaging depend on the degree and duration of the intratumor hemorrhage. Intratumor hemorrhage infrequently induces hepatic rupture and hemoperitoneum (61).

### Marked Hyperintensity on T1-weighted Images

Hyperintensity on T1-weighted images suggests the presence of paramagnetic substances,

**Figure 12.** Liver metastases in a 52-year-old man with mucinous adenocarcinoma of the sigmoid colon. (A) Axial fat-suppressed T2-weighted image shows liver metastases (arrows) with marked hyperintensity due to abundant mucin. (B) Axial portal phase MR image shows liver metastases (arrows) with poor but reticular enhancement.



**Figure 13.** Liver metastasis with a fluid-fluid level in a 61-year-old man with a history of grade 2 NET of the pancreas 1 year previously. (A) Axial arterial phase CT image shows a liver metastasis that appears to be irregularly thickened with an enhancing wall (arrowheads), whereas the central area shows no enhancement. (B) Axial fat-suppressed T2-weighted image shows a fluid-fluid level (dashed arrow) caused by intratumoral hemorrhage. The irregularly thickened wall of the metastasis (arrowheads) is composed of viable tumor tissue and shows moderate hyperintensity. Other metastases (solid arrows) that do not accompany necrosis or degeneration are depicted as areas of moderate hyperintensity.

including melanin (with malignant melanoma), extracellular methemoglobin (with hemorrhage), protein (with ovarian adenocarcinoma, multiple myeloma, and pancreatic mucinous cystic tumors), and necrosis (with colorectal adenocarcinoma). In fact, marked hyperintensity at T1-weighted imaging is characteristic of malignant melanoma (Fig 14A) (45). Other lesions can show mild to slight hyperintensity on T1-weighted images (Fig 6C) (45). Note that marked hyperintensity at T1-weighted imaging can mask the vascularity seen on dynamic contrast-enhanced MR images (Fig 14B). Melanin quantification at T1-weighted imaging is significantly associated with overall survival (62).

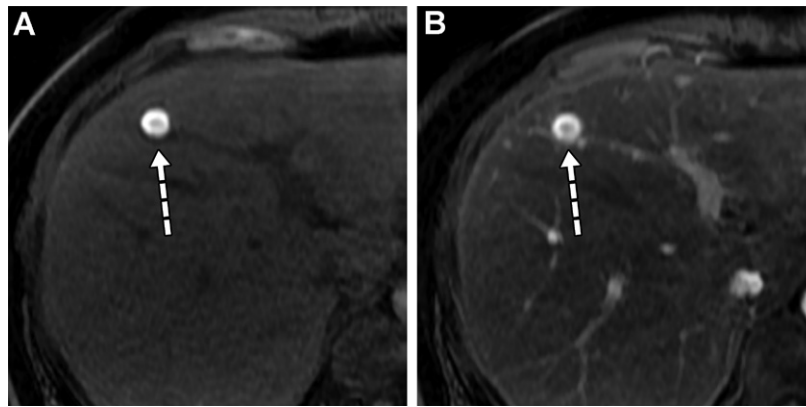
### Calcification

Calcification related to liver metastases is most commonly observed in primary tumors of mucinous adenocarcinoma of the colorectum and stomach, breast, and high-grade serous carci-

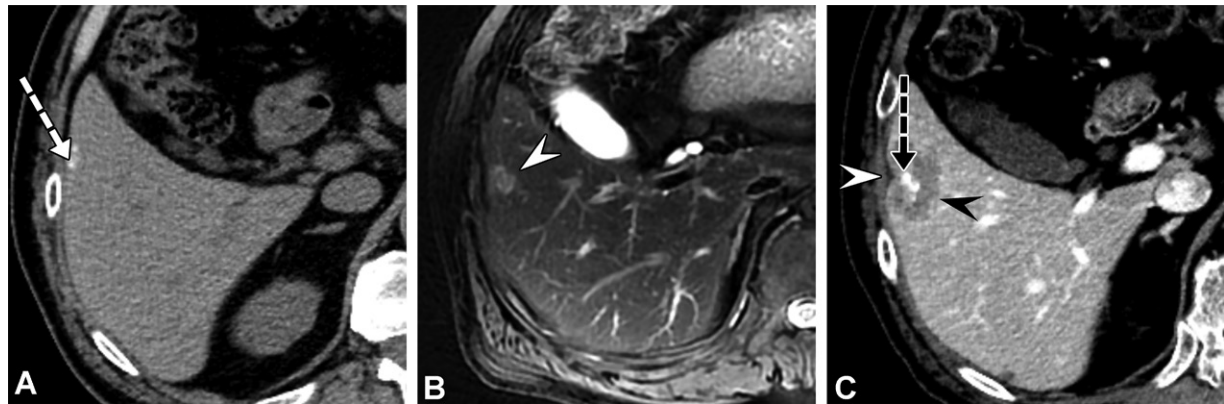
noma, although it is rarely observed in other tumors such as thyroid carcinoma and chondrosarcoma (63). Intratumor calcification is clearly depicted on noncontrast CT images (Fig 15), with patterns and distributions (central and peripheral) that have been found to vary (64).

In addition, the calcification is often accompanied by capsular retraction adjacent to the subcapsular area (Fig 15C). In small metastases, it is possible that only the calcification will be depicted at CT (Fig 15A); therefore, care should be taken not to miss the small area of high attenuation.

Although the development of calcifications is considered to be a marker of treatment response, especially in *KRAS* wild-type colorectal adenocarcinoma treated with antiepidermal growth factor receptor drugs, the correlation between calcification of pretreatment metastases and effectiveness of chemotherapies or prognosis remains controversial (65).



**Figure 14.** Liver metastasis with marked hyperintensity at T1-weighted imaging in a 54-year-old man with malignant melanoma of the submandibular gland who was undergoing chemotherapy. (A) Axial precontrast fat-suppressed T1-weighted image shows a liver metastasis (arrow) with marked hyperintensity due to melanin. (B) Axial arterial phase MR image also shows the metastasis (arrow) with marked hyperintensity.



**Figure 15.** Liver metastasis in an 80-year-old man with a history of prostate and sigmoid adenocarcinomas. (A) Axial noncontrast CT image 6 months after colonic resection shows small areas of calcification (arrow). (B) Follow-up axial fat-suppressed T2-weighted image shows the hyperintense metastasis (arrowhead). (C) Axial portal phase CT image 1 month later shows the hypovascular metastasis (black arrowhead), which has increased in size with increased intratumor calcification (arrow) and capsular retraction (white arrowhead). Given the high association between liver metastases and intratumor calcification, the mass was most likely to be due to metastases from the patient's colon carcinoma rather than from his prostate carcinoma; this was confirmed at surgical-pathologic analysis.

### Hepatic Capsular Retraction

When the convex hepatic contour exhibits focal flattening or concavity, this is defined as hepatic capsular retraction (66). A variety of hepatic neoplasms and nonneoplastic lesions, including liver metastases, have been reported to be associated with hepatic capsular retraction. Capsular retraction usually occurs adjacent to subcapsular metastases (Figs 6, 15, E2) from primary tumors such as lung, breast, and colorectal adenocarcinomas (66). These tumors tend to appear to accompany substantial fibrosis caused by desmoplastic reactions around the metastases (34).

### Vessel-penetrating Sign

The vessel-penetrating sign is an imaging finding in which intrahepatic vessels course through the tumor without disruption or constriction, and it is often observed in primary and secondary lymphomas (Fig E9) (67). Primary hepatic lymphoma is extremely rare, whereas secondary liver involvement occurs in 50% of patients with

non-Hodgkin lymphoma (67,68). In addition to the vessel-penetrating sign, other typical imaging features—such as homogeneous attenuation or signal intensity at CT or MRI, respectively; homogeneous poor enhancement relative to the background liver parenchyma on contrast-enhanced images; and relatively low ADC—could be useful clues for diagnosing lymphoma (27).

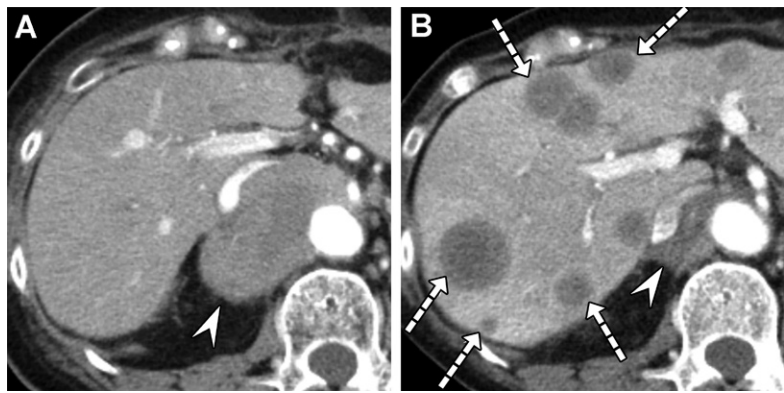
When a multifocal liver mass is encountered without the vessel-penetrating sign at follow-up of a patient with lymphoma, it could suggest another newly appearing malignant tumor with liver metastases (Fig 16). However, this is a rare occurrence.

### Distribution of Liver Metastases

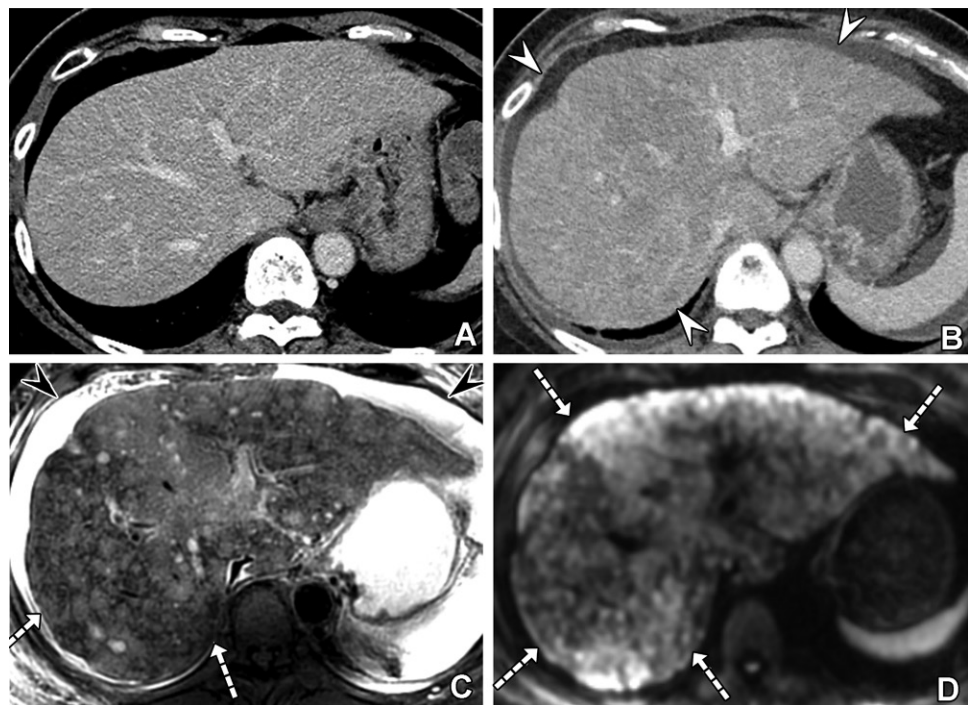
The most common distribution of liver metastases is that of multifocal nodular lesions. However, these metastases sometimes manifest as a solitary mass or confluent masses (Fig E10). In addition, the following two characteristic distributions should be known: diffuse small nodular metastases and metastases prominently located in the gallbladder fossa of the liver.



**Figure 16.** Multiple hepatic masses in a 64-year-old man with follicular lymphoma who had a partial response to chemotherapy. (A) Pretreatment axial arterial phase CT image shows a well-circumscribed hypovascular mass (arrowhead) in the aortocaval space. (B) Follow-up axial portal phase CT image shows multiple new low-attenuation hepatic masses (arrows), which lack the vessel-penetrating sign, and an interval decrease in the size of the lymphoma (arrowhead) after chemotherapy. This finding suggested a new primary malignancy rather than hepatic lymphoma and was proven to be metastatic sarcomatoid carcinoma of the small intestine (not shown).



**Figure 17.** Miliary liver metastases in a 65-year-old woman with a history of invasive ductal carcinoma of the breast 3 years previously. (A) Axial equilibrium phase CT image obtained 6 months previously. (B) Compared with A, the axial equilibrium phase CT image shows diffuse enlargement of the entire liver (arrowheads). (C) Axial fat-suppressed T2-weighted image clearly depicts diffusely spread small metastases (arrows) as hyperintense areas. Perihepatic ascites (arrowheads) also is depicted. (D) Axial diffusion-weighted image clearly depicts diffusely spread small metastases (arrows) as hyperintense areas.



### Diffuse Small Nodular Metastases (Miliary Liver Metastases)

On rare occasions, tumors can infiltrate diffusely and spread as tiny nodules in the whole liver. This phenomenon is referred to as miliary liver metastases and has been reported with lung and breast carcinomas, uveal melanoma, and NET (69). This manifestation can sometimes be accompanied by diffuse intrasinusoidal metastases, which are clearly depicted at MRI as numerous tiny nodular metastases on T2- or diffusion-weighted images (Fig 17). On the other hand, CT images do not depict these tiny metastatic nodules and show only enlargement of the entire liver (Fig 17B).

### Metastases Prominently Located in the Gallbladder Fossa

Gallbladder tumors often metastasize prominently to the gallbladder fossa of the liver as multiple or solitary lesions (Fig 18). This occurs

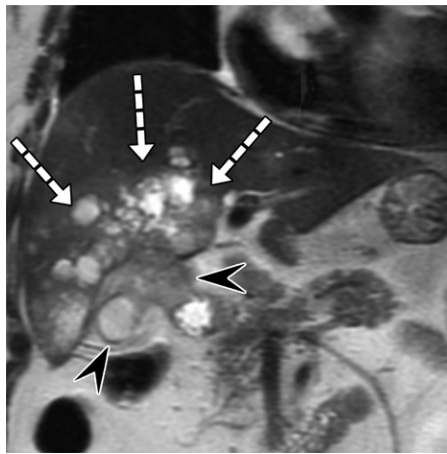
because the drainage veins of the gallbladder (cystic veins) flow into the intrahepatic portal branches or sinusoids around the gallbladder fossa (70).

### Rare Intraductal Forms of Metastases

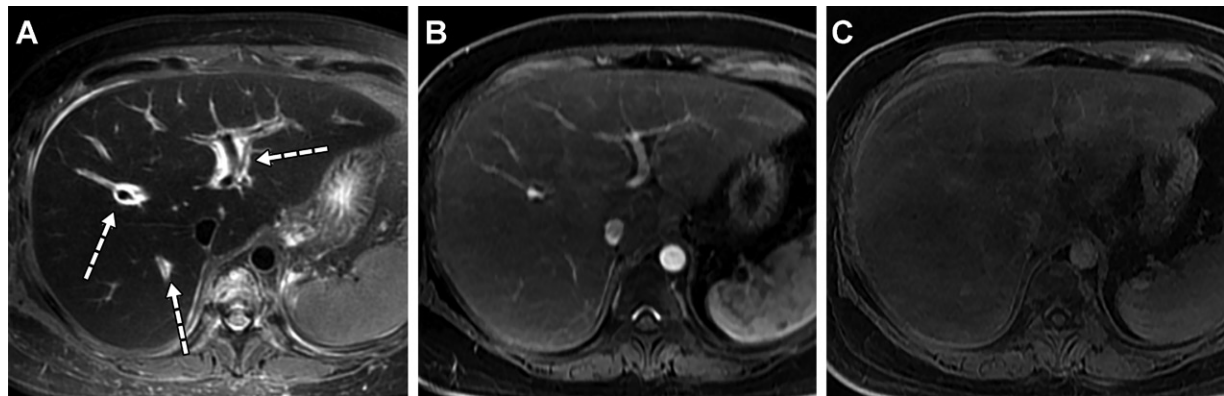
#### Intrasinusoidal Metastases

Most metastases form nodular lesions in the liver parenchyma and rarely appear after ductal metastases. Rarely, tumors infiltrate and spread along the hepatic sinusoids instead of forming discrete masses, making the disease radiologically occult. These rare metastases can be associated with solid tumors, including breast cancer, small cell lung cancer, gastric cancer, and melanoma, and may cause acute liver failure at a high rate (71).

Despite these life-threatening conditions, noncontrast and contrast-enhanced CT and MRI sometimes depict hepatomegaly only (Fig 19).



**Figure 18.** Liver metastases prominently located at the gallbladder fossa in a 72-year-old man with gallbladder adenocarcinoma. Coronal T2-weighted image depicts gallbladder adenocarcinoma (arrowheads) and multiple liver metastases (arrows) prominently located at the gallbladder fossa.



**Figure 19.** Intrasinusoidal liver metastases in a 43-year-old woman who had a history of breast cancer treated with chemotherapy and presented clinically with signs of acute liver failure. (A) Axial fat-suppressed T2-weighted image shows periportal hyperintensity (arrows) but no nodular lesions. (B) Axial arterial phase MR image shows relatively poor enhancement of the liver, which otherwise shows no discrete mass. (C) Axial hepatobiliary phase MR image shows diffuse hypointensity of the entire liver parenchyma, indicating no residual functioning hepatocytes. Intrasinusoidal spread of metastatic breast cancer was histopathologically proven with liver biopsy.

Hepatobiliary phase images can show diffuse hypointensity, which indicates no uptake of gadoxetic acid (Fig 19C). This is because expression of organic anion-transporting polypeptide 1B3 at the sinusoidal membrane, which uptakes gadoxetic acid, would be disturbed by the intrasinusoidal spread of malignant cells (72).

### Portal Vein Tumor Thrombosis

In contrast to the thrombus seen with hepatocellular carcinoma, intrahepatic tumor thrombus occurs infrequently with liver metastases. Involvement of the intrahepatic portal vein has been reported with several tumors, such as gastric and colorectal adenocarcinoma and gastric NET (73). In general, intrahepatic tumor thrombus is usually associated with and shows continuity with nodular metastases (Fig 20).

### Biliary Metastasis

There are two types of biliary metastases: biliary epithelium metastases and intraluminal metastases of the bile duct, and both types can occur

concurrently. Intra- and extrabiliary epithelium metastases rarely occur with colorectal, gastric, breast, uterine, or ovarian carcinomas (74) and involve homogeneous thickening and enhancement of the bile duct, with smooth tapered dilatation of this duct.

Extrabiliary epithelium metastases are typically accompanied by other metastases such as those involving a hepatic parenchymal mass or multiple lymph node metastases (Fig 21). When only the extrahepatic bile duct is involved, it is difficult to differentiate primary bile duct cancer. Furthermore, dilatation of the bile duct can sometimes be indistinguishable from compression of the bile duct by lymph node metastases within the hepatoduodenal ligament.

Intraluminal metastases of the bile duct (ie, intrabiliary metastases) form a polypoid mass within the lumen of the bile duct. The polypoid mass can invade the biliary epithelium and lead to dilatation of the peripheral bile duct, which is often accompanied by a solid metastatic mass in the parenchyma (75). Such metastases have also



**Figure 20.** Portal vein tumor thrombosis in an 80-year-old man with advanced gastric carcinoma. Axial contrast-enhanced CT image shows extensive portal vein thrombus (arrows) through the lateral segment but without a discrete parenchymal mass.



**Figure 21.** Biliary epithelium metastasis in a 64-year-old woman with a history of mucinous adenocarcinoma of the ovary 5 years previously. Axial portal phase CT image shows an intrahepatic bile duct (arrows) with diffuse thickening; this was clinically diagnosed as biliary epithelium metastasis. Multiple lymph node metastases (arrowheads) in the hepatoduodenal ligament also are depicted.

been reported with colorectal adenocarcinoma, melanoma, and hepatocellular carcinoma (75).

Intrabiliary growth metastases can be difficult to distinguish from cholangiocarcinoma; however, a periductal expansile mass has been found to be more common in intrabiliary metastases, whereas a periductal infiltrative lesion has been observed with cholangiocarcinoma (75). Furthermore, intrabiliary metastases can sometimes be indistinguishable from bile duct compression by liver metastases. However, it is more important that dilatation of the bile duct at the periphery of a liver metastasis could be an indicator of poor prognosis (76,77) (Fig E11).

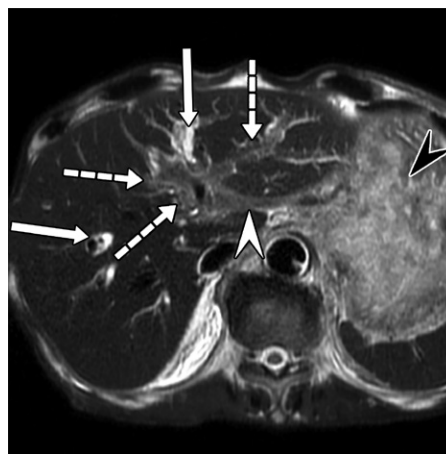
### Lymphatic Invasion

Intrahepatic lymphatic invasion by tumor cells is a manifestation of diffuse lymphatic spread of malignant cells. It has been reported in 4% of cases of liver metastases, especially those of gastric cancer and lymphoma (78). Malignant cells within lymphatic vessels form a periportal mass and dilate the lymphatic vessels.

On CT images, a periportal soft-tissue mass is depicted as a hypoattenuating lesion with attenuation similar to that of water; on T2-weighted images, it is seen as a lesion with moderate hyperintensity. In contrast, dilated lymphatic vessels show periportal hypoattenuation at CT and periportal hyperintensity at T2-weighted (Fig 22) and diffusion-weighted imaging.

### Transformation of Primary Tumor into Different Histopathologic Type

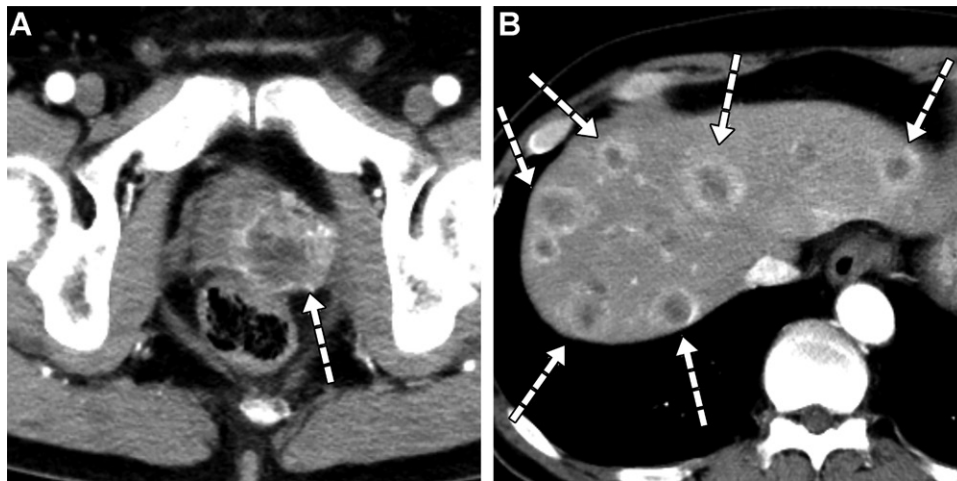
Transformation of a primary cancer into a different histopathologic tumor type has been reported as a resistance process in response to



**Figure 22.** Intrahepatic lymphatic invasion in a 66-year-old man with advanced gastric carcinoma. Axial fat-suppressed T2-weighted image clearly depicts direct invasion of the intrahepatic lymphatic vessels through the hepatogastric ligament (white arrowhead) from the huge gastric carcinoma (black arrowhead). Intrahepatic lymphatic invasion leads to formation of a periportal mass (dashed arrows), which is depicted with mild hypointensity, whereas dilated lymphatic vessels (solid arrows) are depicted with periportal hyperintensity.

targeted therapies; this is closely related to the earlier-mentioned epithelial-to-mesenchymal transition (19,79). For example, histopathologic transformation into small cell lung carcinoma can occur in 3%–15% of patients with non-small cell lung carcinoma, mainly in the setting of adenocarcinoma. In addition, this transformation has been described as a cause





**Figure 23.** Multiple liver metastases in a 68-year-old man after undergoing hormonal therapy. He had prostate adenocarcinoma 3½ years previously. (A) Axial arterial phase CT image shows a mass (arrow) with peripherally dominant enhancement in the left lobe of the prostate. (B) Axial arterial phase CT image shows multiple liver metastases (arrows) with ringlike enhancement. The prostate mass and the multiple hepatic masses suggested hypervascular metastases with necrosis or degeneration at imaging and were proven to be NET transdifferentiated from adenocarcinoma.

of acquired tyrosine kinase inhibitor resistance (80). Treatment-related neuroendocrine prostate cancer transdifferentiates from adenocarcinoma as an adaptive response to androgen receptor pathway inhibition (Fig 23) (81).

Although transformation into aggressive lymphomas such as diffuse large B-cell lymphoma or Burkitt lymphoma also is known to occur with indolent lymphomas such as follicular lymphoma, marginal zone lymphoma, and lymphoplasmacytic lymphoma, it has been best described as occurring with follicular lymphoma (82). Transformation of a primary cancer into a different histopathologic type can also affect the imaging findings, reflecting histologic features, with the liver metastasis also exhibiting an appearance similar to that of the primary change.

### Imaging Biomarkers of Prognosis or Treatment Response

Several imaging findings described in earlier sections of this review could be used for predicting the prognosis or the effectiveness of chemotherapies, serving as imaging biomarkers. However, it is necessary to understand that these results are not absolute and could change in the future on the basis of data from larger studies.

### Conclusion

Although the imaging features of liver metastases from various primary malignancies overlap, obtaining a better understanding of the mechanisms that underlie organ-specific metastases and determining the characteristic features of liver metastases from specific primary cancers can help provide a more precise diagnosis.

### References

1. Horn SR, Stoltzfus KC, Lehrer EJ, et al. Epidemiology of liver metastases. *Cancer Epidemiol* 2020;67:101760.
2. Ros PR, Taylor HM. Malignant tumors of the liver. In: Gore RM, Levine MS, eds. *Textbook of gastrointestinal radiology*. 2nd ed. Philadelphia, Pa: Saunders, 2000; 1523–1568.
3. Massarweh NN, El-Serag HB. Epidemiology of Hepatocellular Carcinoma and Intrahepatic Cholangiocarcinoma. *Cancer Contr* 2017;24(3):1073274817729245.
4. Kemeny N, Kemeny M, Dawson L. Liver Metastases. In: Shimm DS, ed. *Abeloff's Clinical Oncology*. 4th ed. London, England: Elsevier, 2008; 846–862.
5. Hess KR, Varadhachary GR, Taylor SH, et al. Metastatic patterns in adenocarcinoma. *Cancer* 2006;106(7):1624–1633.
6. Park JH, Kim JH. Pathologic differential diagnosis of metastatic carcinoma in the liver. *Clin Mol Hepatol* 2019;25(1):12–20.
7. Winter J, Auer RAC. Metastatic malignant liver tumors. In: Jarnagin WR, Blumgart LH, eds. *Blumgart's Surgery of the Liver, Pancreas and Biliary Tract*. 6th ed. Philadelphia, Pa: Elsevier, 2017; 1290–1304.
8. Ewing J. *Neoplastic diseases: a treatise on tumors*. 3rd ed. Philadelphia, Pa: Saunders, 1928; 1049–1051.
9. Wilkinson AL, Qurashi M, Shetty S. The Role of Sinusoidal Endothelial Cells in the Axis of Inflammation and Cancer Within the Liver. *Front Physiol* 2020;11:990.
10. Paget S. The distribution of secondary growths in cancer of the breast. 1889. *Cancer Metastasis Rev* 1989;8(2):98–101.
11. Bakalian S, Marshall JC, Logan P, et al. Molecular pathways mediating liver metastasis in patients with uveal melanoma. *Clin Cancer Res* 2008;14(4):951–956.
12. Brodt P. Role of the Microenvironment in Liver Metastasis: From Pre- to Prometastatic Niches. *Clin Cancer Res* 2016;22(24):5971–5982.
13. Ruivo CF, Adem B, Silva M, Melo SA. The Biology of Cancer Exosomes: Insights and New Perspectives. *Cancer Res* 2017;77(23):6480–6488.
14. Kalluri R, LeBleu VS. The biology, function, and biomedical applications of exosomes. *Science* 2020;367(6478):eaau6977.
15. Hoshino A, Costa-Silva B, Shen TL, et al. Tumour exosome integrins determine organotropic metastasis. *Nature* 2015;527(7578):329–335.
16. Liu Y, Cao X. Organotropic metastasis: role of tumor exosomes. *Cell Res* 2016;26(2):149–150.
17. Hapach LA, Mosier JA, Wang W, Reinhart-King CA. Engineered models to parse apart the metastatic cascade. *NPJ Precis Oncol* 2019;3(1):20.

18. Fares J, Fares MY, Khachfe HH, Salhab HA, Fares Y. Molecular principles of metastasis: a hallmark of cancer revisited. *Signal Transduct Target Ther* 2020;5(1):28.
19. Diepenbruck M, Christofori G. Epithelial-mesenchymal transition (EMT) and metastasis: yes, no, maybe? *Curr Opin Cell Biol* 2016;43:7–13.
20. Choi SH, Kim SY, Park SH, et al. Diagnostic performance of CT, gadoxetate disodium-enhanced MRI, and PET/CT for the diagnosis of colorectal liver metastasis: systematic review and meta-analysis. *J Magn Reson Imaging* 2018;47(5):1237–1250.
21. Choi JY, Choi JS, Kim MJ, et al. Detection of hepatic hypovascular metastases: 3D gradient echo MRI using a hepatobiliary contrast agent. *J Magn Reson Imaging* 2010;31(3):571–578.
22. Kim HJ, Lee SS, Byun JH, et al. Incremental value of liver MR imaging in patients with potentially curable colorectal hepatic metastasis detected at CT: a prospective comparison of diffusion-weighted imaging, gadoxetic acid-enhanced MR imaging, and a combination of both MR techniques. *Radiology* 2015;274(3):712–722.
23. Schulz A, Viktil E, Godt JC, et al. Diagnostic performance of CT, MRI and PET/CT in patients with suspected colorectal liver metastases: the superiority of MRI. *Acta Radiol* 2016;57(9):1040–1048.
24. Mungai F, Morone M, Villanacci A, et al. Diffusion weighted MR and apparent diffusion coefficient measurement in classification and characterization of noncystic focal liver lesions: does a clinical role exist? *Medicine (Baltimore)* 2014;93(6):e40.
25. Sutherland T, Steele E, van Tonder F, Yap K. Solid focal liver lesion characterisation with apparent diffusion coefficient ratios. *J Med Imaging Radiat Oncol* 2014;58(1):32–37.
26. Gultekin MA, Turk HM, Yurtsever I, et al. Apparent Diffusion Coefficient Values for Neuroendocrine Liver Metastases. *Acad Radiol* 2021;28(suppl 1):S81–S86.
27. Colagrande S, Calistri L, Grazzini G, et al. MRI features of primary hepatic lymphoma. *Abdom Radiol (NY)* 2018;43(9):2277–2287.
28. Lee JE, Kim SH, Lee S, Choi SY, Hwang JA, Woo SY. Differentiating metastatic mucinous colorectal adenocarcinomas from simple cysts of the liver using contrast-enhanced and diffusion-weighted MRI. *Br J Radiol* 2018;91(1091):20180303.
29. Jackson T, Darwish M, Cho E, Nagatomo K, Osman H, Jeyarajah DR. 68Ga-DOTATATE PET/CT compared to standard imaging in metastatic neuroendocrine tumors: a more sensitive test to detect liver metastasis? *Abdom Radiol (NY)* 2021;46(7):3179–3183.
30. Breedis C, Young G. The blood supply of neoplasms in the liver. *Am J Pathol* 1954;30(5):969–977.
31. Namasivayam S, Martin DR, Saini S. Imaging of liver metastases: MRI. *Cancer Imaging* 2007;7(1):2–9.
32. Lincke T, Zech CJ. Liver metastases: detection and staging. *Eur J Radiol* 2017;97:76–82.
33. Eble JA, Niland S. The extracellular matrix in tumor progression and metastasis. *Clin Exp Metastasis* 2019;36(3):171–198.
34. Whatcott CJ, Diep CH, Jiang P, et al. Desmoplasia in primary tumors and metastatic lesions of pancreatic cancer. *Clin Cancer Res* 2015;21(15):3561–3568.
35. Henke E, Nandigama R, Ergün S. Extracellular Matrix in the Tumor Microenvironment and Its Impact on Cancer Therapy. *Front Mol Biosci* 2020;6:160.
36. Semelka RC, Hussain SM, Marcos HB, Woosley JT. Perilesional enhancement of hepatic metastases: correlation between MR imaging and histopathologic findings—initial observations. *Radiology* 2000;215(1):89–94.
37. van Dam PJ, van der Stok EP, Teuwen LA, et al. International consensus guidelines for scoring the histopathological growth patterns of liver metastasis. *Br J Cancer* 2017;117(10):1427–1441.
38. Frentzas S, Simoneau E, Bridgeman VL, et al. Vessel co-option mediates resistance to anti-angiogenic therapy in liver metastases. *Nat Med* 2016;22(11):1294–1302.
39. Calistri L, Rastrelli V, Nardi C, et al. Imaging of the chemotherapy-induced hepatic damage: yellow liver, blue liver, and pseudocirrhosis. *World J Gastroenterol* 2021;27(46):7866–7893.
40. Lee SL, Chang ED, Na SJ, et al. Pseudocirrhosis of breast cancer metastases to the liver treated by chemotherapy. *Cancer Res Treat* 2014;46(1):98–103.
41. Ozaki K, Kozaka K, Kosaka Y, Kimura H, Gabata T. Morphometric changes and imaging findings of diffuse liver disease in relation to intrahepatic hemodynamics. *Jpn J Radiol* 2020;38(9):833–852.
42. Park YS, Lee CH, Kim BH, et al. Using Gd-EOB-DTPA-enhanced 3-T MRI for the differentiation of infiltrative hepatocellular carcinoma and focal confluent fibrosis in liver cirrhosis. *Magn Reson Imaging* 2013;31(7):1137–1142.
43. Fleming M, Ravula S, Tatishchev SF, Wang HL. Colorectal carcinoma: pathologic aspects. *J Gastrointest Oncol* 2012;3(3):153–173.
44. Sica GT, Ji H, Ros PR. Computed tomography and magnetic resonance imaging of hepatic metastases. *Clin Liver Dis* 2002;6(1):165–179, vii.
45. Kelekis NL, Semelka RC, Woosley JT. Malignant lesions of the liver with high signal intensity on T1-weighted MR images. *J Magn Reson Imaging* 1996;6(2):291–294.
46. Outwater E, Tomaszewski JE, Daly JM, Kressel HY. Hepatic colorectal metastases: correlation of MR imaging and pathologic appearance. *Radiology* 1991;180(2):327–332.
47. Mahfouz AE, Hamm B, Wolf KJ. Peripheral washout: a sign of malignancy on dynamic gadolinium-enhanced MR images of focal liver lesions. *Radiology* 1994;190(1):49–52.
48. Ha S, Lee CH, Kim BH, et al. Paradoxical uptake of Gd-EOB-DTPA on the hepatobiliary phase in the evaluation of hepatic metastasis from breast cancer: is the “target sign” a common finding? *Magn Reson Imaging* 2012;30(8):1083–1090.
49. Fujita N, Nishie A, Asayama Y, et al. Hyperintense Liver Masses at Hepatobiliary Phase Gadaxetic Acid-enhanced MRI: Imaging Appearances and Clinical Importance. *RadioGraphics* 2020;40(1):72–94.
50. Park SH, Kim H, Kim EK, et al. Aberrant expression of OATP1B3 in colorectal cancer liver metastases and its clinical implication on gadoxetic acid-enhanced MRI. *Oncotarget* 2017;8(41):71012–71023.
51. Lee HJ, Lee CH, Kim JW, Park YS, Lee J, Kim KA. Use of hepatobiliary phase images in Gd-EOB-DTPA-enhanced MRI of breast cancer hepatic metastasis to predict response to chemotherapy. *Clin Imaging* 2017;43:127–131.
52. Arnason T, Fleming KE, Wanless IR. Peritumoral hyperplasia of the liver: a response to portal vein invasion by hypervascular neoplasms. *Histopathology* 2013;62(3):458–464.
53. Fujita N, Handa A, Park JM. Peritumoral Hyperintensity at Hepatobiliary Phase Gadaxetic Acid-enhanced MRI in Hepatic Neuroendocrine Tumors. *Anticancer Res* 2022;42(2):911–917.
54. Silva AC, Evans JM, McCullough AE, Jatoi MA, Vargas HE, Hara AK. MR imaging of hypervascular liver masses: a review of current techniques. *RadioGraphics* 2009;29(2):385–402.
55. Tang M, Li Y, Lin Z, et al. Hepatic nodules with arterial phase hyperenhancement and washout on enhanced computed tomography/magnetic resonance imaging: how to avoid pitfalls. *Abdom Radiol (NY)* 2020;45(11):3730–3742.
56. Stafford-Johnson DB, Francis IR, Eckhauser FE, Knol JA, Chang AE. Dual-phase helical CT of nonfunctioning islet cell tumors. *J Comput Assist Tomogr* 1998;22(2):335–339.
57. Qian LJ, Zhu J, Zhuang ZG, Xia Q, Liu Q, Xu JR. Spectrum of multilocular cystic hepatic lesions: CT and MR imaging findings with pathologic correlation. *RadioGraphics* 2013;33(5):1419–1433.
58. Matsushita N, Douhara A, Ueno H, et al. Intraductal papillary-mucinous carcinoma with portal vein tumor thrombus and multifocal liver metastasis: an autopsy case. *Mol Clin Oncol* 2021;14(5):101.
59. Lee JE, Kim SH, Lee S, Choi SY, Hwang JA, Woo SY. Differentiating metastatic mucinous colorectal adenocarcinomas from simple cysts of the liver using contrast-enhanced and diffusion-weighted MRI. *Br J Radiol* 2018;91(1091):20180303.
60. Sommer WH, Zech CJ, Bamberg F, et al. Fluid-fluid level in hepatic metastases: a characteristic sign of metastases of neuroendocrine origin. *Eur J Radiol* 2012;81(9):2127–2132.

61. Schoedel KE, Dekker A. Hemoperitoneum in the setting of metastatic cancer to the liver: a report of two cases with review of the literature. *Dig Dis Sci* 1992;37(1):153–154.
62. Ramtohul T, Ait Rais K, Gardrat S, et al. Prognostic Implications of MRI Melanin Quantification and Cytogenetic Abnormalities in Liver Metastases of Uveal Melanoma. *Cancers (Basel)* 2021;13(11):2728.
63. Patnana M, Menias CO, Pickhardt PJ, et al. Liver calcifications and calcified liver masses: pattern recognition approach on CT. *AJR Am J Roentgenol* 2018;211(1):76–86.
64. Yu MH, Kim YJ, Park HS, Jung SL, Jeon HJ. Imaging patterns of intratumoral calcification in the abdominopelvic cavity. *Korean J Radiol* 2017;18(2):323–335.
65. Zhou Y, Zhang J, Dan Pu, et al. Tumor calcification as a prognostic factor in cetuximab plus chemotherapy-treated patients with metastatic colorectal cancer. *Anticancer Drugs* 2019;30(2):195–200.
66. Blachar A, Federle MP, Sosna J. Liver lesions with hepatic capsular retraction. *Semin Ultrasound CT MR* 2009;30(5):426–435.
67. Rajesh S, Bansal K, Sureka B, Patidar Y, Bihari C, Arora A. The imaging conundrum of hepatic lymphoma revisited. *Insights Imaging* 2015;6(6):679–692.
68. Ross A, Friedman LS. The liver in systemic disease. In: Bacon BR, O'Grady JG, Di Bisceglie AM, Lake JR, eds. *Comprehensive clinical hepatology*. 2nd ed. Philadelphia, Pa: Mosby Elsevier, 2006; 537–547.
69. Mishima S, Nozaki Y, Mikami S, et al. Diffuse liver metastasis of small-cell lung cancer presenting as acute liver failure and diagnosed by transjugular liver biopsy: a rare case in whom nodular lesions were detected by enhanced CT examination. *Case Rep Gastroenterol* 2015;9(1):81–87.
70. Sugita M, Ryu M, Satake M, et al. Intrahepatic inflow areas of the drainage vein of the gallbladder: analysis by angio-CT. *Surgery* 2000;128(3):417–421.
71. Ogata T, Kikawa Y, Ogata M, Satake H, Hatachi Y, Yasui H. Acute Liver Failure with Diffuse Liver Metastasis from Breast Cancer, Not Detected by Computed Tomography: 2 Case Reports. *Case Rep Oncol* 2018;11(3):699–704.
72. Leonhardt M, Keiser M, Oswald S, et al. Hepatic uptake of the magnetic resonance imaging contrast agent Gd-EOB-DTPA: role of human organic anion transporters. *Drug Metab Dispos* 2010;38(7):1024–1028.
73. Tomimaru Y, Sasaki Y, Yamada T, et al. Liver metastasis originating from colorectal cancer with macroscopic portal vein tumor thrombosis: a case report and review of the literature. *J Med Case Reports* 2010;4(1):382.
74. Min JH, Jang KM, Cha DI, et al. Differences in early imaging features and pattern of progression on CT between intrahepatic biliary metastasis of colorectal origin and intrahepatic non-mass-forming cholangiocarcinoma in patients with extrabiliary malignancy. *Abdom Radiol (NY)* 2019;44(4):1350–1360.
75. Lee YJ, Kim SH, Lee JY, et al. Differential CT features of intraductal biliary metastasis and double primary intraductal polypoid cholangiocarcinoma in patients with a history of extrabiliary malignancy. *AJR Am J Roentgenol* 2009;193(4):1061–1069.
76. Koh FH, Shi W, Tan KK. Biliary metastasis in colorectal cancer confers a poor prognosis: case study of 5 consecutive patients. *Ann Hepatobiliary Pancreat Surg* 2017;21(1):57–60.
77. Nakai Y, Gonoi W, Kurokawa R, et al. MRI findings of liver parenchyma peripheral to colorectal liver metastasis: a potential predictor of long-term prognosis. *Radiology* 2020;297(3):584–594.
78. Pupulin LF, Vilgrain V, Ronot M, Becker CD, Breguet R, Terraz S. Hepatic lymphatics: anatomy and related diseases. *Abdom Imaging* 2015;40(6):1997–2011.
79. Zheng X, Carstens JL, Kim J, et al. Epithelial-to-mesenchymal transition is dispensable for metastasis but induces chemoresistance in pancreatic cancer. *Nature* 2015;527(7579):525–530.
80. Dorantes-Heredia R, Ruiz-Morales JM, Cano-García F. Histopathological transformation to small-cell lung carcinoma in non-small cell lung carcinoma tumors. *Transl Lung Cancer Res* 2016;5(4):401–412.
81. Puca L, Vlachostergios PJ, Beltran H. Neuroendocrine Differentiation in Prostate Cancer: Emerging Biology, Models, and Therapies. *Cold Spring Harb Perspect Med* 2019;9(2):a030593.
82. Montoto S, Fitzgibbon J. Transformation of indolent B-cell lymphomas. *J Clin Oncol* 2011;29(14):1827–1834.